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TABLE 2
FILLING MODE

Sensor Position	Condition	LED1	LED2	LED3	PUMP
Water touches no sensors (below lowest sensor)	Tank is empty	OFF	OFF	OFF	ON
Water touches the lowest sensor	Tank is not empty	ON	OFF	OFF	ON
Water touches the middle sensor	Tank is partially filled	OFF	ON	OFF	ON
Water touches the highest sensor	Tank is full	OFF	OFF	ON	OFF

TABLE 3
DRAINING MODE

Sensor Position	Condition	LED1	LED2	LED3	PUMP
Water level is at the highest sensor	Tank is full	OFF	OFF	ON	OFF
Water level is below the highest sensor	Tank is partially filled	OFF	ON	OFF	OFF
Water level is below the middle sensor	Tank is not empty	ON	OFF	OFF	OFF
Water level is below the lowest sensor	Tank is empty	OFF	OFF	OFF	ON

heights in the tank. The pump motor starts automatically when water is below the lowest sensor.

When water touches the lowest sensor, LED1 lights up, but the pump remains on. When water touches the middle sensor, LED1 turns off, LED2 lights up, and the pump remains on.

When water touches the highest sensor, LED3 lights up, LEDs 1 and 2 turn off, and the pump switches off.

Draining mode

During water drainage, when the water level drops below the highest sensor, LED3 turns off, LED2 lights up, but the pump still remains off.

When the water level reaches below the middle sensor, LED2 turns off, LED1 lights up, but the pump remains off.

When the water is below the lowest sensor, all LEDs turn off, and the pump starts.

Construction and testing

The source code must first be uploaded into the Arduino Uno. The components should then be connected as per the circuit diagram shown in Fig. 3. The Arduino Uno board should be powered using a 5V adaptor or a USB cable connected to a laptop or desktop.

The water pump should be connected to a 230V AC power supply through the relay contacts. After powering the device and installing the sensors in the tank, water should be filled into the tank. The LEDs will glow according to the water level, and the pump will switch on or off as described in Tables 2 and 3. **EFY**

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